

Arika Khor

SWE & Researcher with Interest in Autonomous Systems, Digital Facial Privacy Protection, and System Design

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Education

Expected B.S. 2026 , M.S. 2027	The University of Oklahoma B.S. / M.S. Computer Science, focus in Robotics & Computer Vision
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Experience

June 2026 - Current	The Boeing Company <i>Intern: Simulation Software Engineering</i> · Worked with podman containers for on-device builds of previous on-prem server only codebases. · Developed a hybrid Retrieval-Augmented Generation + spec-driven model which: – Used several sql lite DBS storing: mapped codebase, context history, project phases, document summaries. – Delegated subagents to speed up initial tasks such as codebase mapping and PDF reading. · Helped build gitlab build pipelines on podman containers · Refactored out stale GNU Autotools build from Real Time System codebase · Wrote Software Design Documents
Sept 2022 - July 2024	Singapore Armed Forces <i>Curriculum Development Data Engineer</i> · Prediction based data analysis on Cadet scores using pandas, simple linear regression, and · Migrated excel sheets used in trainee scoring to more modern processing with python (todo)
Jan 2020 - Dec 2020	GameDuchy Game Development Company <i>Assistant Community Relations Manager</i> · Planned, organised and executed regular in-game community wide events, working with various departments to successfully implement these events. · Designed and implemented custom chat bot to obtain item data.

Research Labs

June 2025 - Current	Construction Innovation and Analytics Lab (CinnaLAB) <i>Time to Incident Predictor on a Autonomous vehicle in dynamic construction sites</i> · Implmented Autonomous 4-wheel differential drive robot navigating on a known dynamic construction site. · Used Nav2, ROS2, and · Sensor stack: LIDAR (Unitree L1), Depth camera (zed2i), GNSS (sparkfun GPS-RTK)
Aug 2025 - Current	Autonomous Interactive Robotics Lab (AIROU) <i>Collaborative MARL Autonomous Vehicles at Smart Intersections in Nvidia Isaac Lab</i> · MARL Enviornment · RGB only self driving car · Implemented a smart intersection using a traffic controller with Multi-agent reinforcement learning robots · <i>ARCPRO Autonomous Model Race Cars</i> · Created Ansible images which handled network and dependency for a collection of mini PCs · Created website to host all documentation on alongside build process · Setup software stack for race cars, ...

Leadership

Aug 2024 - Current	Sooner Competitive Robotics <i>Software Member & Treasurer (25-26)</i> · Designed new UI for autonomous robot with: – Real time playback of ROS bags – Switch modes and cameras
Aug 2016 - May 2022	Scouting America <i>Various Leadership Positions</i> · Eagle Scout Project: Designed and built several weather radio antennas by organizing a large group to mass-produce them. They were then given to the city's emergency coordinator to distribute to public service buildings. This project was covered in the normantranscript (article here).
Aug 2024 - Current	Sooner Competitive Programming Team <i>Team Capitan (24-26, Team Advisor (26-27)</i>

Notable Projects

2025 Hacklahoma

Facial-Net Classifier

Facial Recognition Project

- Fully local real time Facial Recognition suite to raise awareness about academia and policy making.
- Captures live video then detects and matches faces in a crowd attempting to either match or generate a new face with mongodb.
- Used quartz 4 and obsidan for documentation.
- UI to merge/delete/assign saved faces using Streamlit and flask.

Technical Skills

Strong Skills

- Tailscale, LaTeX, Java, Docker, Git, Linux (Debian)

General Skills

- Programming: Advanced Java (Spring,Thymeleaf), C+ +, C#, Java Script, Python, Bash
- Web: LaTeX, HTML5, CSS3, Bootstrap, Azure, AWS, SQL, MonogDB, Bun JS, React, WebRTC Concepts
- Tools & Concepts : ROS2, Pub/Sub Architecture, Docker, Jira, Jenkins , Git, Microsoft Office Suite, Agile, Scrum, tailscale.
- Soft Skills: Leadership, Interdisciplinary Teamwork, Communication, Problem Solving, Project management.
- OS and Machines: Debian, Windows, Jetson AGX Orin (Jetpack 6.2)